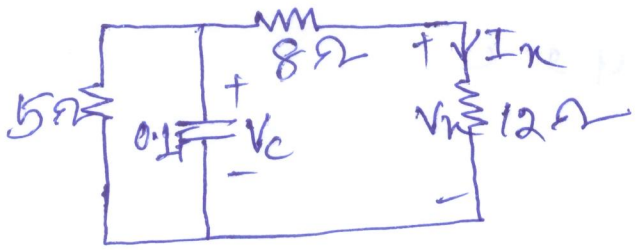
Lecture-10

Pb No-1 Find V_c, V_x & I_x for $t>0$, Let $V_c(0) = 15V$



Ans: $R_{th} = \frac{5\times(8+12)}{5+(8+12)} = \frac{5\times 20}{25} = 4\,\Omega$

$\tau = RC = 4\times 0.1 = 0.4$ sec

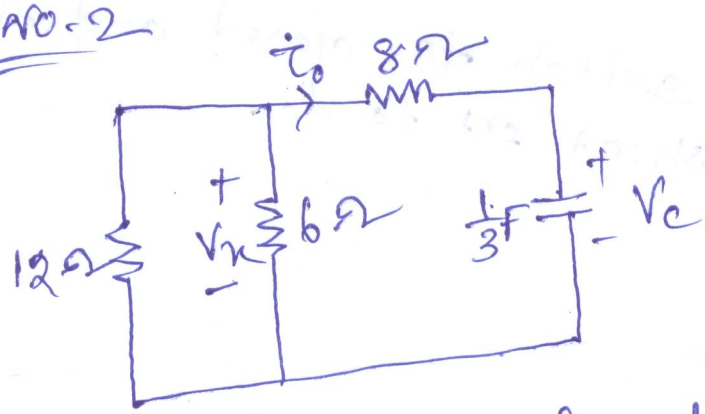
At $t>0$, capacitor act as open ckt

So $V_c(t) = V_c(0)\, e^{-\frac{t}{\tau}} = 15\, e^{-\frac{t}{0.4}} = 15\, e^{-2.5t}$ volt

$I_x = \frac{V_c(t)}{8+12} = \frac{15}{20} e^{-2.5t}$ Amp

$V_x = I_x \times 12 = \frac{3}{\cancel{12}\times\frac{\cancel{15}}{\cancel{20}}} e^{-2.5t} = 9\, e^{-2.5t}$

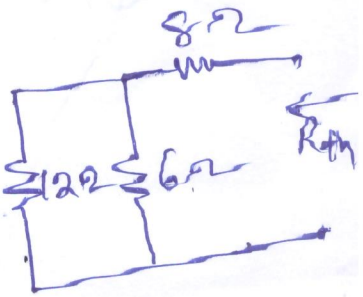
Pb No-2



Find V_c, V_x & i_0 for $t>0$, Given $V_c(0) = 60V$

Ans: $R_{th_1} = \frac{12\times 6}{12+6} = \frac{\cancel{72}^{4}}{\cancel{18}_{3}} = 4\,\Omega$

$R_{th} = R_{th_1} + 8 = 12\,\Omega$



$$V_c(t) = V_c(0) \times e^{-\frac{t}{\tau}} = 60\, e^{-\frac{t}{4}} = 60\, e^{-0.25t} \text{ volt}$$

$$\tau = R_{th} C = 12 \times \frac{1}{3} = 4 \text{ sec}$$

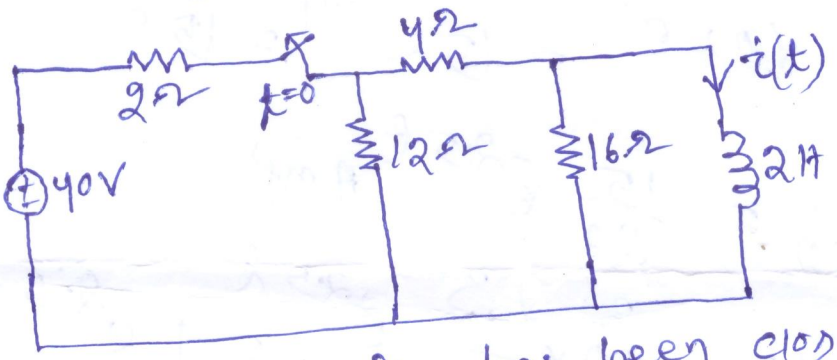
KCL at node 'A' :-

$$\frac{V}{12} + \frac{V}{6} + \frac{V - V_c(t)}{8} = 0$$

$$\Rightarrow V_x = 20\, e^{-0.25t} \text{ volt}$$

$$i_0 = \frac{V_x - V_c}{8} = \frac{-40}{8} e^{-0.25t} = 5 e^{-0.25t} \text{ A}$$

Pb-3



The switch in the fig has been closed for a long time. At $t=0$, the switch is opened. calculate $i(t), t > 0$.

Ans: when $t < 0$, the switch is closed and the inductor act as short ckt to dc.

$$i(0) = 6\text{A}, \quad t < 0$$

$$i(t) = i(0)\, e^{-4t}$$